

Recommendation **ITU-T K.113 (07/2025)**

SERIES K: Protection against interference

Generation of radiofrequency electromagnetic field level maps

Recommendation ITU-T K.113

Generation of radiofrequency electromagnetic field level maps

Summary

Recommendation ITU-T K.113 provides guidance on how to make radiofrequency electromagnetic field (RF-EMF) maps for assessing existing exposure levels over large areas of cities or territories and for an appropriate public disclosure of the results, in a simple and understandable way.

History*

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Introduction

One of the problems for deployment of telecommunication infrastructure, with regard to public opinion, is the distrust generated by the general population's lack of knowledge of electromagnetic field emissions and their resulting exposure levels.

In some countries, this distrust inhibits the deployment of these critical infrastructures, causing unnecessary tensions among the parties involved: citizens, governments and telecommunication operators.

This Recommendation provides a practical tool for assessing exposure levels in large areas of cities or territories, which complements the simulations, specific measurements and continuous monitoring already being carried out in many countries. (Technical guidance can be found in [ITU-T K.52], [ITU-T K.61], [ITU-T K.70], [ITU-T K.83], [ITU-T K.91] and [ITU-T K.100].) This Recommendation describes methods and tools for the development of radiofrequency electromagnetic field (RF-EMF) maps.

This Recommendation also describes methods for communicating RF-EMF exposure data in a visual manner, i.e., through exposure maps, which may be better understood by the general public.

Examples of national adaptations are included.

Recommendation ITU-T K.113

Generation of radiofrequency electromagnetic field level maps

1 Scope

This Recommendation describes methods and characteristics of the systems used for generating radiofrequency electromagnetic field (RF-EMF) maps to assess and communicate RF-EMF exposure levels that people can expect to find in certain areas.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

- [[ITU-T K.52](#)] Recommendation ITU-T K.52 (2024), *Guidance on complying with limits for human exposure to electromagnetic fields*.
- [[ITU-T K.61](#)] Recommendation ITU-T K.61 (2018), *Guidance on measurement and numerical prediction of electromagnetic fields for compliance with human exposure limits for telecommunication installations*.
- [[ITU-T K.70](#)] Recommendation ITU-T K.70 (2020), *Mitigation techniques to limit human exposure to EMFs in the vicinity of radiocommunication stations*.
- [[ITU-T K.83](#)] Recommendation ITU-T K.83 (2024), *Monitoring of electromagnetic field levels*.
- [[ITU-T K.91](#)] Recommendation ITU-T K.91 (2024), *Guidance for assessment, evaluation and monitoring of human exposure to radio frequency electromagnetic fields*.
- [[ITU-T K.100](#)] Recommendation ITU-T K.100 (2024), *Measurement of radio frequency electromagnetic fields to determine compliance with human exposure limits when a base station is put into operation*.
- [ISO/IEC 17025] ISO/IEC 17025 (2005), *General requirements for the competence of testing and calibration laboratories*.
- [ISO/IEC Guide 98-3] ISO/IEC Guide 98-3 (2008), *Uncertainty of measurement – Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)*.
- [EN 50413] Cenelec EN 50413 (2019), *Basic standard on measurement and calculation procedures for human exposure to electric, magnetic and electromagnetic fields (0 Hz – 300 GHz)*.
- [IEC 60529] IEC 60529 (1989), *Degrees of protection provided by enclosures (IP Code)*.
- [IEC 62232] IEC 62232:2025, *Determination of RF field strength, power density and SAR in the vicinity of base stations for the purpose of evaluating human exposure*.

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 antenna [ITU-T K.70].

3.1.2 assessment [IEC 62232]: Undertaking of an evaluation of "RF exposure from an EUT" in order to arrive at a judgement about comparison with the applicable exposure limits.

3.1.3 averaging time [ITU-T K.52].

3.1.4 base station (BS) [ITU-T K.100].

3.1.5 electric field strength (E) [ITU-T K.83].

3.1.6 electromagnetic field (EMF) [ITU-T K.91].

3.1.7 evaluation, <of an exposure metric> [IEC 62232]: Process of determining a value "of an exposure metric".

3.1.8 exposure [ITU-T K.83].

3.1.9 exposure level [ITU-T K.52].

3.1.10 exposure limits [ITU-T K.70].

3.1.11 far-field region [ITU-T K.83].

3.1.12 general public [ITU-T K.52].

3.1.13 IP (ingress or international protection) [IEC 60529]: The IP Code, International Protection Marking, sometimes interpreted as Ingress Protection Marking, classifies and rates the degree of protection provided against intrusion (body parts such as hands and fingers), dust, accidental contact, and water by mechanical casings and electrical enclosures.

3.1.14 linearity [ITU-T K.83].

3.1.15 magnetic field strength (H) [ITU-T K.83].

3.1.16 near-field region [ITU-T K.83].

3.1.17 power density (S) [ITU-T K.83].

3.1.18 root mean square (rms) [ITU-T K.83].

3.1.19 user equipment (UE) [b-IEC TR 62669]: Device used directly by an end user to communicate in a mobile network.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

3.2.1 frequency selective measurement system: A field-quantity measuring instrument capable of receiving and processing the spectral component of a particular emission within the instrument response frequency range.

3.2.2 measurement period: The time between the beginning of a measurement and the start of the next measurement.

3.2.3 measurement time: The time that measuring equipment needs to perform each measurement, i.e., to take each sample.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

5G	fifth Generation
BS	Base Station
CW	Continuous Wave
EMF	Electromagnetic Field
GPS	Global Positioning System
RF	Radiofrequency
RF-EMF	Radiofrequency Electromagnetic Field
rms	root mean square

5 Conventions

None.

6 Methods

Three possible methods to generate RF-EMF maps are described in clauses 6.1 to 6.3; however, other methods providing similar information may also be used.

6.1 Drive test measurements

RF-EMF maps can be generated according to the drive test method, which is widely known in the telecommunication sector, as it is regularly employed to assess coverage and other related telecommunication network parameters.

The drive test method consists of continuously collecting metrics from a moving vehicle.

Traffic rules shall be complied with and driving safety ensured when undertaking drive test measurements.

The vehicle is equipped with measurement equipment and a positioning system.

Positioning system equipment is needed to enable geographical and simultaneous positioning of the measurements.

Measurement equipment may also respond to up-link radiofrequency (RF) signals from mobile phones and other radio transmitters. Unless frequency-selective measurements are taken, measurement staff should not use mobile phones and other radio transmitters.

Using this method, the following data may be obtained for each point:

[Position] [date/time] [sample value]

Since the measurements are taken on the move, the measurement time (or integration time) of the equipment used should be considered (see the definition for measurement time in clause 3.2.3). It is recommended that samples (measurements) be completed within approximately 5 m for urban areas and 10 m for non-urban areas. However, distances between measurements should be decided depending on the mapping purpose.

Therefore, the maximum allowed speed of the vehicle must be calculated according to the measurement time of the equipment being used.

Example 1: measuring equipment with a measurement time of 300 ms:

- if the vehicle moving at 50 km/h, it will require 4.2 m for each sample to be taken;

- the maximum speed would be 60 km/h in order to stay within the 5 m distance between samples (urban area).

Example 2: measuring equipment with a measurement time of 1 s:

- the maximum speed to stay within the limit of 5 m between samples is 18 km/h (urban areas);
- the maximum speed to stay within the limit of 10 m between samples is 36 km/h (non-urban areas).

Given that most common GPS devices provide position information every second, a measurement period of 1 s is recommended (see the definition for measurement period in clause 3.2.2).

In any case, speed requires adaptation to the state of the road, since measurement sensors can generate erroneous readings due to sudden movements in the presence of static fields.

The movement of the vehicle may not permit time averaged measurements as recommended by [b-ICNIRP-1] or [b-ICNIRP-2] and other national and international organizations. However, it does give an approximation of the RF-EMF levels over large areas that otherwise would be impossible to cover. To overcome this limitation, the vehicle should stop at least every hour to get a statistically valid sample of specific averaged measurements to be taken as a reference. These samples of averaged measurements are compared with the measurements taken in motion, to ensure that the order of magnitude is similar.

In all cases, the system checks that all measurements taken in motion are at least 6 dB below the most restrictive limit of the entire frequency band being measured with the probe. If the value of any measurement is above this threshold, the measuring equipment is moved to that point to perform specific time averaged measurements of the unperturbed root mean square (rms) field (with a non-conductive tripod and the operator keeping an appropriate distance).

Example:

- probe for measuring EMFs in the frequency range 100 kHz – 8 GHz (see clause 7);
- most restrictive reference level for the general public in this band according to [b-ICNIRP-2]: 27.7 V/m (or 2 W/m²);
- acceptance threshold (6 dB under the most restrictive limit): 14 V/m (0.5 W/m²) (to simplify).

Therefore, when measuring in the 100 kHz to 8 GHz band, all values below the acceptance threshold are considered adequate, and values above the acceptance threshold are specifically re-measured with a 6-minute average.

NOTE – 100 kHz to 8 GHz is recommended as it covers broadcast, mobile communication FR1, and Wi-Fi bands (up to 7.125 MHz).

6.2 Theoretical calculations

Alternatively, RF-EMF maps can be constructed by theoretical calculations. In this case, it is essential to carry out periodic tests in a representative sample of locations averaged over six or thirty minutes, or to demonstrate through other techniques that calculated values are representative of actual measured levels. Technical guidance can be found in [ITU-T K.61], [ITU-T K.70], [ITU-T K.91], [ITU-T K.100] and [IEC 62232].

Check that the calculated levels do not underestimate the measurement levels, taking into account environmental and other factors (e.g., call traffic on mobile networks).

6.3 Grid method

Measurements may be taken in cities using the following procedure:

- Check the geographical size of the city/area.

- The assessment area is divided into multiple square grids in advance. The invalid grids are those where the area cannot be accessed (such as water surface, restricted area) or the area is built-up on less than 50% of its surface.
- The dimension of the square grids will depend on the purpose of the mapping.
- Take the coordinates of the spots located at the vertices of the grid. If the grid point is not suitable for measurement (such as water surface, restricted area, road lane, etc.), the point should be adjusted to the nearest available measurable position, and the maximum allowable distance between the adjusted and the original point is 1/4 of the grid length. Additional measurement points can be added to the grid according to requirements.
- At each selected point on the grid, measure the RF-EMFs according to international standards (like [ITU-T K.100] or [IEC 62232]). Shorter time averaging periods may be acceptable (in accordance with [ITU-T K.100]) to save time.
- For points where limit is exceeded, the individual in charge must check both the spot and the surroundings, meeting the recommendations of [b-ICNIRP-1] or [b-ICNIRP-1] and [b-ICNIRP-3] or the national limits and the specific measuring methodology already established in [ITU-T K.52], [ITU-T K.61] [ITU-T K.100], [IEC 62232] or in applicable national standards).

Mobile phones should not be used by measurement staff during a measurement.

With the information obtained from each of the measured points (geographic coordinates, date/time, the average value of the sample taken), the results should be plotted on the corresponding RF-EMF level maps.

6.4 Combined method

A method that combines two or three of the methods in clauses 6.1 to 6.3 or any other appropriate technique may also be applied. For example, a combined method could use theoretical calculations in some areas and drive tests in others.

6.5 Discrepancies between results

Regardless of the method, if there is a discrepancy between results, data obtained through time-average measurements (either with broadband or frequency-selective measurement system) are to prevail.

7 Specifications of the equipment to be used

7.1 Equipment for the drive test

The system to be installed in the vehicle consists of:

- 1) Electromagnetic isotropic field probe;
- 2) GPS;
- 3) Protection of measurement equipment from environmental conditions (see clause 7.2.3.4);
- 4) Optional: laptop and UE.

7.1.1 Electromagnetic isotropic field probe

Every time a sample is taken, the full value of the electric field strength (E) should be obtained. In order to avoid errors, the reading of the three axes of the broadband isotropic field probe (composed of E or H antennas) should preferably be simultaneous.

The frequency band of the probe should be appropriate for the frequencies to be measured. Such probes provide a flat response in very large frequency ranges. In many cases, a response between

100 kHz and 8 GHz is sufficient, but alternatively probes that provide broader responses or narrower responses can be used if the contributions for particular RF services are of interest.

The probe should take rms measurements.

Other important features of this type of probe are:

- Measurement range: minimum rms value 0.3-20 V/m; minimum continuous wave (CW) value 0.3-100 V/m;
- Sensitivity: minimum 0.3 V/m;
(NOTE – This is approximately equal to the arithmetic mean reported in European cities [b-Urbinello]).
- Frequency response: maximum ± 3 dB in the band under consideration;
- Linearity: maximum ± 1 dB;
- Isotropy: maximum ± 2.5 dB up to 3 GHz (± 3.5 dB up to 6 GHz);
- Temperature range: -10°C to 45°C .

The position of the field probe on the vehicle should be between 1.5 and 3 m above ground level.

7.2 Equipment for grid test or spot measurements

Spot measurements refer to all measurements made with the vehicle stopped (clause 6.1), measurements to compare with theoretical calculation (clause 6.2) and grid method measurements (clause 6.3). Either broadband or frequency-selective equipment can be used.

7.2.1 Broadband equipment

Broadband equipment permits the total value of the field in the frequency band of interest to be obtained and the results to be directly compared with those obtained by theoretical calculation or the drive test. Broadband equipment mounted on vehicles can also calculate averaged measurements, in which case it is only necessary to stop the vehicle during the six minutes needed to obtain the desired measurement.

Otherwise, the measurement should be performed using a tripod and then included in the file. The values measured by broadband probe are compared with the lower limit of the working band of the probe, unless the probe has a shaped frequency response. If the value is more than 6 dB below this limit, the result is valid, otherwise it is necessary to repeat the measurements using frequency-selective equipment.

The recommended height for spot measurements taken with equipment not mounted on the vehicle is 1.5 m.

7.2.2 Frequency-selective equipment

Frequency-selective equipment permits specific values for each telecommunication service to be obtained. The sum of the contributions should be calculated in order to obtain the total RF-EMF exposure value and include the results in the file. The sum of the contributions must be made as specified by [b-ICNIRP 1998] or [b-ICNIRP 2020] (the latter is described in Appendix I of [ITU-T K.91]).

7.2.3 Data acquisition electronics

Electronics should enable data capture by the probe and storage of each piece of information with its date and time, plus its geographical position.

7.2.3.1 Measurement system

The working performance of the measurement system should meet the requirements of RF-EMF to be measured as provided for in [ITU-T K.52], [ITU-T K.61] [ITU-T K.100], [IEC 62232] or in

applicable national standards. The measurement instrument produces readings as rms, time averaged as required by applicable standards. The measurement instrument also meets the following requirements.

- During measurement, a frequency-selective RF-EMF measurement instrument is used; the measurement frequency is selected as the downlink frequency band when the BS transmitting antenna under test is working.
- When measuring RF-EMF exposure from 5G and other network system BSs at the same site, a frequency-selective RF-EMF measurement instrument has the means to conveniently measure different frequency ranges in a sequence.
- When measuring, the measurement system uses an isotropic probe, which meets requirements in Table 1. If a non-isotropic probe is used, the influence of antenna directivity is considered and the antenna factor and other parameters are synthesized in result processing; the detection direction and polarization is adjusted in the measurement until the maximum field intensity is measured.
- To support the instrument, a tripod constructed of non-conductive materials that is not susceptible to moisture is used.

NOTE – When 5G is being tested, measurement procedures should be followed as specified in ITU-T Recommendations (K.61, K.91, K.100) or IEC 62232.

7.2.3.2 Measurement instrument

Table 1 lists the basic specifications of a frequency-selective measurement instrument.

**Table 1 – Basic specifications of a frequency-selective measurement instrument
(see [IEC 62232])**

Item	Specification
Frequency response	900 MHz~3 GHz, $\leq \pm 1.5$ dB
	< 900 MHz or > 3 GHz, $\leq \pm 3$ dB
Dynamic range	> 60 dB
Probe detection limit	The lower detection limit of the probe is $< 7 \times 10^{-6}$ W/m ² (0.05 V/m) The upper detection limit was ≥ 25 W/m ² (100 V/m)
Linearity	$\leq \pm 1.5$ dB
Frequency error	$< 10^{-3}$ Orders of magnitude of the measured frequency
Isotropic	< 900 MHz, isotropic < 2 dB 900 MHz~3 GHz, isotropic < 3 dB > 3 GHz, isotropic < 5 dB

7.2.3.3 GPS

GPS permits geographical location data to be obtained at any time via the satellite network. It can be an external element of the system or it can be integrated into the continuous measurement equipment, to directly associate the measurement data at each spot with the geographical position.

7.2.3.4 Environmental protection

Having to always work outdoors, a minimum IP55 environmental protection level [IEC 60529] is recommended.

7.2.3.5 Laptop and user equipment (UE) (optional)

The direct connection between the measuring equipment and the GPS (if it is external) permits data to be monitored and visualized in real time and the evolution of the drive test to be checked by suitable mapping.

Note especially that measurements of in-situ RF-EMF exposures for 5G and 3G/4G differ. For a 5G BS, at least one item of 5G user equipment (UE) should be connected to the BS under test and measured when there is downlink traffic. Alternative approaches may be specified in national requirements.

Figure 1 is a simple layout of the in-situ RF-EMF exposure measurement for a 5G BS site.

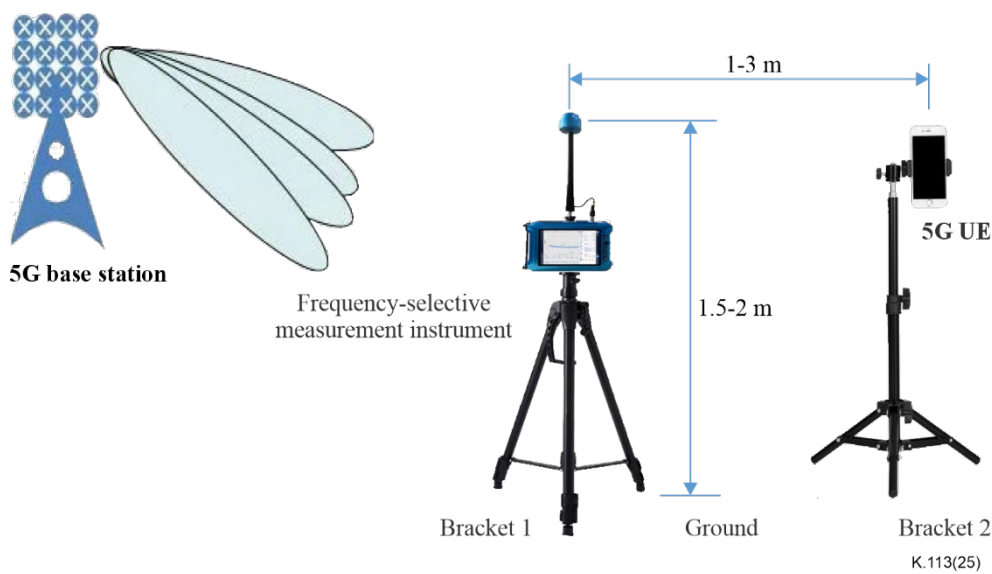


Figure 1 – In-situ RF-EMF exposure measurement layout of a 5G BS

Figure 2 shows UE for the in-situ RF-EMF exposure measurement for a 5G BS site.



Figure 2 – UE for in-situ RF-EMF exposure measurement

A) Environmental conditions for measurement

Weather conditions, e.g., rain, temperature, meet the operating requirements of the measurement instrument.

B) Measurement factor

The measured variable of the RF-EMF environment of a BS is RF-EMF, and the metric is the power density.

C) Measurement point selection

In the case of a 5G measurement, the measurement point should be arranged in the coverage range of the beams of the 5G BS antenna.

7.3 Calibration

All measuring equipment should be individually calibrated by an accredited ISO/IEC 17025 laboratory. It is advisable to make accredited calibrations and calibrations recognized by the International Laboratory Accreditation Cooperation (ILAC).

8 Uncertainties

The uncertainties shall be estimated in compliance with methods described in [EN 50413], [ITU-T K.91], [IEC62232] and [ISO/IEC Guide 98-3]. The contributions to the total uncertainty of the measurement can be obtained through the appropriate measurements performed on the equipment or according to the manufacturer's specifications. Normally, tolerances with rectangular distribution can be considered.

The expanded uncertainty with a confidence interval of 95 per cent shall not exceed 4 dB.

The contributions of each component of uncertainty shall be registered with their name, probability distribution, sensitivity coefficient and uncertainty value. The results shall be recorded as described in Table 2. The combined uncertainty shall then be evaluated according to equation 1:

$$u_c = \sqrt{\sum_{i=1}^m c_i^2 u_i^2} \quad (1)$$

where c_i is the weighting coefficient (sensitivity coefficient). The expanded uncertainty shall be evaluated using a confidence interval of 95 per cent which can be obtained with coverage factor $k = 1.96$ for a near-normal distribution, typical in most measurements.

Table 2 gives a practical guideline for assessing the uncertainty in measurements of EMFs.

There may be other uncertainties not listed in Table 2. If the parties concerned find other relevant uncertainties, these should be added and taken into account for the overall calculation.

Table 2 – Uncertainty assessment

Error sources	Uncertainty value % uv_i	Probability distribution	Divisor k_i	c_i	Standard uncertainty % $u_i = uv_i/k_i$
Measurement equipment					
Calibration		Normal	2	1	
Isotropy		Rectangular	$\sqrt{3}$	1	
Linearity		Rectangular	$\sqrt{3}$	1	
Frequency response		Rectangular	$\sqrt{3}$	1	
Environmental parameters					
Influence of temperature and humidity in the measuring equipment		Rectangular	$\sqrt{3}$	1	
Influence of the vehicle		Rectangular	$\sqrt{3}$	1	
Post-processing					
Combined standard uncertainty					
Expanded uncertainty (confidence interval of 95%)		Normal			$u_e = 1.96 \cdot u_c$

9 Report

The ultimate goal of RF-EMF maps is to relay information on RF-EMF levels to the public in an understandable visual format.

RF-EMF levels are plotted based on the total number of samples and in relation to the percentage of the national limits. It is advisable to use a colour code to reflect the levels of the different spots on the maps.

The colour code should always be part of the map. A map without a colour code may not be understandable and may lead to misunderstandings. An example of colour code can be found in Appendix II.

NOTE – Appendix I also suggests an alternative monochrome approach for displaying results.

The map shall also be accompanied by the following information:

- Measurement or calculation location;
- Date;
- Description of the system or equipment used, or of calculation tools;
- Calibration record of the equipment used;
- Measurement or calculation uncertainty;
- Entity in charge of making the RF-EMF maps;
- Scale and limit values.

Appendix I

Example of colour code and alternatives to use of colours for displaying RF-EMF map levels

(This appendix does not form an integral part of this Recommendation.)

I.1 Example of colour code

Table I.1 is an example of a unified colour code for better understanding and comparison among maps performed in different countries or by different agencies using the same RF-EMF limit values.

This colour code may not be appropriate for people with red-green colour blindness.

Table I.1 – Example of colour code for displaying RF-EMF map levels

Colour										
Name	Maya blue	Dodger blue	Cerulean blue	Light green	Lime green	Green	Golden Yellow	Orange	Orange red	Red
Hexadecimal colour code	#73c2fb	#1e90ff	#2a52be	#90ee90	#32cd32	#008000	#ffd000	#ffa500	#ff4500	#ff0000
Percentage (<i>P</i>) in relation to the E-field limit %	$P \leq 1$	$1 < P \leq 2$	$2 < P \leq 4$	$4 < P \leq 8$	$8 < P \leq 15$	$15 < P \leq 20$	$20 < P \leq 35$	$35 < P \leq 50$	$50 < P \leq 100$	$P > 100$

I.2 Alternative example of colour code

It may be that users of this Recommendation are affected by colour blindness. Colour blindness derives from different sensitivities in the light cones of the human eye and varies in level and quality by individual person. For this reason, it is difficult to recommend a set of colours that can work well for all possibly affected people.

Alternative schemes can be used, e.g., employing monochrome representations of levels with different patterns. Two possible ways are identified in Figure I.1, for the same levels as those of Table I.1. The specification of a recommended monochrome pattern scheme is for further study.

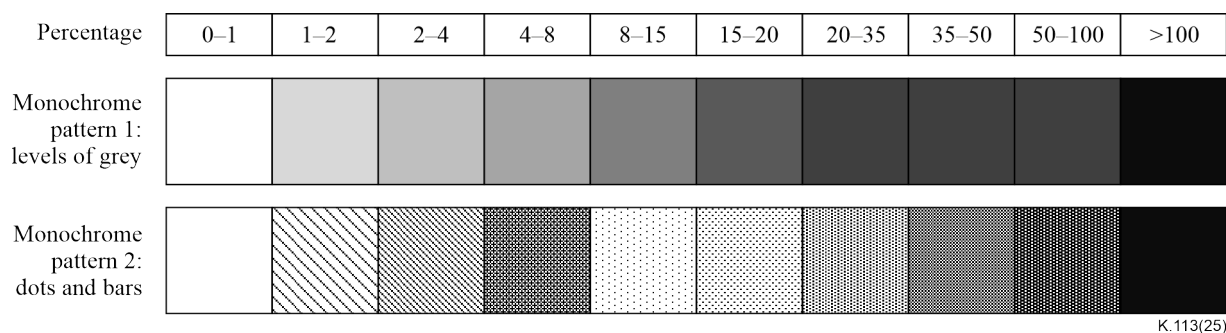


Figure I.1 – Monochrome alternative to colour codes for displaying RF-EMF map levels

Appendix II

Case study on mapping measurements of electromagnetic fields from radiocommunication stations in Brazil

(This appendix does not form an integral part of this Recommendation.)

II.1 Legislation in Brazil

In Brazil, human exposure to electric, magnetic and electromagnetic fields is regulated by Federal Law No. 11,934 of 5 May 2009.

This law establishes that the limits recommended by the World Health Organization (WHO) will be adopted for occupational and general population exposure to electric, magnetic and electromagnetic fields generated by radio transmitter stations operating up to 300 GHz. And until new recommendations are established by WHO, the limits of the International Commission on Non-Ionizing Radiation Protection (ICNIRP), recommended by WHO, will be adopted.

The National Telecommunications Agency (ANATEL) inspects compliance with the limits for human exposure to electric, magnetic and electromagnetic fields generated by radio transmitting stations.

ANATEL was created in 1997 by Federal Law No. 9,472, of 16 July 1997, and in spite of the federal law N° 11,934 entered into force in 2009, in 2002, ANATEL edited the Resolution No. 303/2002 (<https://informacoes.anatel.gov.br/legislacao/resolucoes/17-2002/128-resolucao-303>), which dealt with limiting human exposure to electric, magnetic and electromagnetic fields in the radio frequency band between 9 kHz and 300 GHz, with limits based on the recommendation ICNIRP 1998.

In 2019, Resolution No. 303/2002 was repealed by Resolution No. 700 of 28 September 2018 (<https://informacoes.anatel.gov.br/legislacao/resolucoes/2018/1161-resolucao-700>), which established the regulation on the assessment of human exposure to electric, magnetic and electromagnetic fields associated with the operation of radio transmitting stations with the principles and regulatory policy commands within the limits established in federal legislation.

On the other hand, the technical aspects were approved by the superintendence responsible for administering the use of the radio spectrum through Technical Requirements Act No. 458/2019 (<https://informacoes.anatel.gov.br/legislacao/atos-de-requisitos-tecnicos-de-gestao-do-espectro/2019/1237-ato-458>), which came into force on 30 January 2019, the same date as Resolution No. 700/2018.

With the update of the recommendation ICNIRP 2020, the study to update Act No. 458/2019 began, culminating in the approval of Technical Requirements Act No. 17,865, of 30 December 2023 (<https://informacoes.anatel.gov.br/legislacao/atos-de-requisitos-tecnicos-de-gestao-do-espectro/2023/1914-ato-17865>) with the recommendations of the current ICNIRP limits.

In order to standardize ANATEL's verification of compliance with the limits of human exposure to electric fields, Portaria (internal document) No. 2,768 of 29 January 2024 was approved, approving the Inspection Instruction for assessing exposure to radiofrequency electric, magnetic and electromagnetic fields (EMF) (<https://informacoes.anatel.gov.br/legislacao/instrucoes-de-fiscalizacao/1917-portaria-2768>).

Assessment through measurements can be originated by the following stakeholders: i) public authorities; ii) the general population; and iii) ANATEL's internal requesters.

The public authorities that make the most demands in this area are the Public Prosecutor's Office at state level; the general population makes demands through the "Anatel Consumidor" mobile app and the internal requester is the area responsible for administering the spectrum.

The region or location to be assessed is defined as: i) large areas; ii) small areas; and iii) internal areas.

Large areas are neighbourhoods or municipalities, small areas are spaces such as squares or sports courts and internal areas are spaces inside buildings such as shopping malls, schools, hospitals or buildings.

In large areas, mobile probes will preferably be used to assess the entire area.

In small areas, where it is not possible to use mobile probes for the assessment, portable or fixed probes must be used.

In indoor areas, portable or fixed meters must be used.

Fixed probes will be used in cases where measurements are required at long intervals, such as more than 1 hour.

When planning measurements using mobile probes, priority will be given to areas with a higher concentration of radio transmitting stations and a greater impact on the general population.

For example, there are places such as hills where there is a concentration of stations, but the general population does not regularly access this region.

When evaluating portable or fixed probes, identify possible points with higher emission levels, taking into account the licensed stations in the region. Attention should also be paid to metallic structures, glazing with metallic films and water surfaces that can act as reflectors in an attempt to maintain distance from these conditions.

Instantaneous measurements are important in order to determine points with the highest level of exposure to EMF, and then carry out temporal averages.

For mobile probe measurements, Recommendation ITU-T K.113 is used as a basis.

Meanwhile, for portable probe measurements, [ITU-T K.61] is used as the basis for spatial averaging.

II.2 Flow for assessment

II.2.1 Assessment with mobile probe

Initially, the measurement is carried out using a mobile probe, and if the measurement is less than 14 V/m (Recommendation ITU-T K.113), the assessment in the region is considered compliant.

Otherwise, at the point of measurement, a simplified assessment will be carried out with a portable probe.

II.2.2 Simplified portable probe assessment

The simplified assessment consists of taking measurements at 1.7 m for at least 6 minutes with a portable probe.

If the assessment with a simplified portable probe is less than 27.7 V/m, the assessment result is compliant.

Otherwise, it is necessary to carry out the detailed assessment by performing the spatial average according to [ITU-T K.61] with 3 (three points) (heights of 1.1 m / 1.5 m / 1.7 m).

II.2.3 Detailed portable probe evaluation

The detailed assessment consists of taking measurements with a portable probe, using spatial averaging for at least 6 minutes at each point. Initially, it is carried out at three points (heights of 1.1 m / 1.5 m / 1.7 m), and if the average of the measurements is greater than 27.7 V/m it is necessary to carry it out at 6 or 9 points according to Figure 3.b of [ITU-T K.61].

If the 6 or 9 measurements are greater than 27.7 V/m, it is necessary to carry out the assessment with a narrow-band probe.

II.2.4 Frequency-selective evaluation

The frequency-selective assessment consists of taking measurements with frequency-selective probes.

It is carried out when the 9-point detailed assessment is greater than 27.7 V/m.

The range to be assessed is where the sources individually contribute to a total exposure ratio (TER) of more than 5%, measuring for at least 6 minutes using the same procedures as the simplified portable probe assessment.

It should be noted that each range has its own limit and the most restrictive limit of the range assessed should be adopted.

II.2.5 Evaluation of frequencies below 30 MHz

When assessing individual sources with a TER greater than 5% and emitting frequencies below 30 MHz, assessments must be carried out using electric field probes and magnetic field probes.

The most restrictive limit for magnetic field strength (H) is 0.073 A/m.

II.3 Adoption of precautionary action

Once the assessment has found that the levels are above the limit, there is a risk to human health and precautionary action must be taken, starting with the station with the greatest contribution.

Precautionary action must be taken to bring human exposure into line with the limits for the general population establish in national environment. Initially, the contribution of each emitting source is reduced proportionally until the contribution of all sources is below the limit.

And finally, even if after the exposure reductions the measures are still above the limit for the general population, it may be necessary to carry out precautionary interruption of the emitting source.

Every year, measurements are taken to assess human exposure to RF-EMF in large areas and the measurement data are entered into the ANATEL system, which is available to the public.

The user can select some parameters such as state, municipality, year, so that the values measured at the location of interest are displayed.

Figure II.1 shows an example of selecting the state of the Federal District.

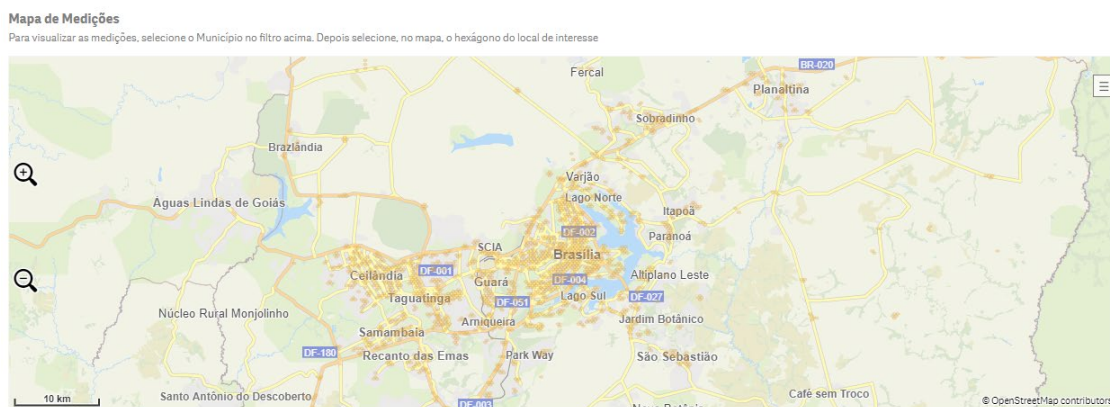


Figure II.1 – Map of exposure to electromagnetic fields in the Federal District of Brazil

Considering the region of interest, you can see that the area has been divided into hexagons and the highest value and the quantity of measurements are displayed, as shown in Figure II.2.

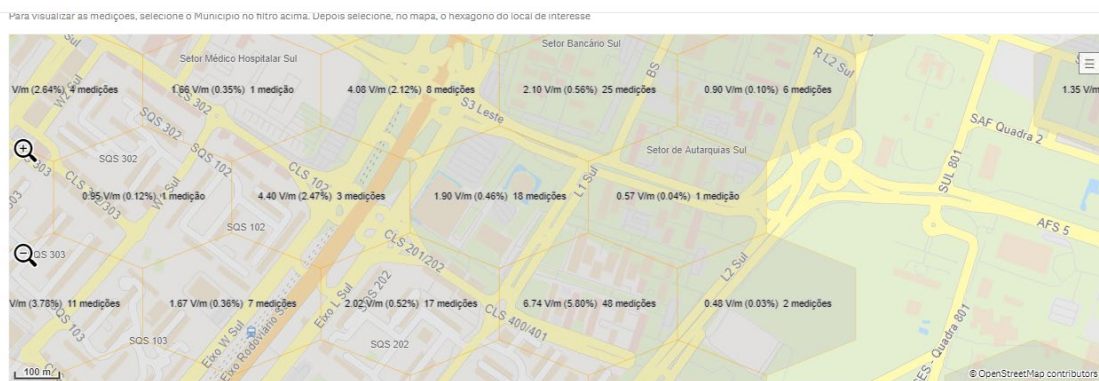


Figure II.2 – Region divided into hexagons and showing the highest value and the quantity of measurements

II.4 Conclusion

This map showing the EMF measurements to the public not only complies with the commands of Federal Law No. 11,934/2009, it also complies with the principle of transparency and demonstrates to society that the values are below the established limits and there is no risk to human health.

This map is constantly being improved to make it more accessible to users who want to know more about human exposure to EMF.

Appendix III

Example of a RF-EMF field level map for the general public in the Republic of Korea

(This appendix does not form an integral part of this Recommendation.)

III.1 RF-EMF field level map in the Republic of Korea

In the Republic of Korea, under the Radio Waves Act, an electromagnetic field (EMF) measurement system is implemented to assess the RF-EMF strength for base stations (BSs). The EMF measurement procedure and methods for mobile BSs are defined in the Korean standard [b-KS C 3367], which is based on [IEC 62232]. Based on the results, the Republic of Korea classifies BSs into four categories according to an EMF grading system. Table III.1 shows the classification criteria used in the Republic of Korea to assign grades to mobile BSs based on the EMF measurement results.

Table III.1 – RF-EMF strength grading system in the Republic of Korea

1 First Grade	2 Second Grade	3 Caution Grade	4 Warning Grade
Less than 50% of the reference level for general public	More than 50% and less than 100% of the reference level for general public	Exceeding the reference level for general public and less than the reference level for occupational	Exceeding the reference level for occupational

Additionally, the Republic of Korea operates an RF-EMF level map to provide RF-EMF measurement results of mobile BSs for the general public. This map displays RF-EMF measurement results using four different colours, based on the RF-EMF grading system mentioned above. An example of how measurement information is presented to the public through this RF-EMF level map is shown in Figure III.1 and Figure III.2. The RF-EMF level map provides not only the colour-coded RF-EMF measurement results of mobile BSs, but also additional information such as the measured RF-EMF values, the mobile BS's operator and the EMF measurement agency.



Figure III.1 – Example of displaying EMF measurement results for a grade 1 mobile BS



Figure III.2 – Example of displaying EMF measurement results for a grade 2 mobile BS

III.2 Simplified example of colour code

The Republic of Korea imposes a legal obligation to measure the RF-EMF strength for mobile BSs. The RF-EMF measurement procedure and methods for mobile BSs are defined in the Korean standard [b-KS C 3367], which is based on [IEC 62232]. Table III.2 shows a simplified four-level colour code designed to help the general public more easily and intuitively understand the RF-EMF measurement results of mobile BSs when using the RF-EMF level map. For example, if the RF-EMF level is less than 50% of the reference level for the general public, it is marked green. However, as with the colour code mentioned in Appendix I, this simplified colour code may not be appropriate for people with red–green colour blindness.

Table III.2 – Example of a colour code for displaying RF-EMF map levels

Colour				
Name	Green	Blue	Orange	Red
Hexadecimal colour code	#32cd32	#1e90ff	#ffa500	#ff0000
Percentage (P) in relation to the E-field limit %	$P \leq 50$ (for general public)	50 (for general public) < $P \leq 100$ (for general public)	100 (for general public) < $P \leq 100$ (for occupational)	100 (for occupational) < P

Appendix IV

Example of RF-EMF field level monitoring for the public in China

(This appendix does not form an integral part of this Recommendation.)

IV.1 Introduction

Regional RF-EMF environmental monitoring systems have been established in difference provinces of the People's Republic of China under the supervision and guidance of local governments. These can be used by authorities to provide information to the public. Chinese mobile network operators collect RF-EMF exposure level data, which are used to generate the maps using methods described in [b-GB 8702].

The standard uses the following terms:

- **Electric field strength percentage of standard (E%)**
The ratio of the square of the electric field strength to the square of the corresponding limit. It is calculated according to the formula of the evaluation method in [b-GB 8702] and expressed as a percentage.
- **Electromagnetic quality index (EMQI)**
Used to characterize and evaluate the electromagnetic environment quality of an area. The electromagnetic quality index is divided into the local area electromagnetic quality Index and the Overall electromagnetic quality index. The local area electromagnetic quality index is the arithmetic mean of the effective measurement point within a specific square grid in the evaluation area, denoted EMQI local; the Overall electromagnetic quality index is the arithmetic mean of all local area electromagnetic quality indicates within the evaluation area, denoted EMQI overall.
- **Assessment area**
According to the regional scope required for electromagnetic quality assessment, this is usually composed of one or several urban built-up areas in administrative regions above the county level.

IV.2 Measurement locations

When measuring outside a building, the measurement point is located at points of interest of the RF-EMF within the coverage range of the BS antenna, such as: areas with public activities in the direction of its main lobe, and the nearest place where the public lives, works or studies from the antenna. However, measurement points are not selected in locations that need to be reached by means of tools (such as ladders) or special means (such as climbing).

When measuring inside a building, the measurement point can be selected in a window (balcony) position towards the BS antenna, and the probe tip of the measuring instrument is within the window frame (balcony) interface, or other indoor positions. In addition, the distance between the probe of the measuring instrument and the equipment such as household appliances is ≥ 1 m.

IV.3 Measurement height

The probe of the measurement instrument is 1.5 m to 2.0 m above the ground or surface. Other measurement heights can also be selected according to different purposes and indicated in the measurement report.

IV.4 Measurement of working conditions and 5G user equipment

During measurement, the 5G BS under test operates under normal working conditions, 5G user equipment (UE) is connected to the 5G BS in at least one of the typical application scenarios

mentioned in the form described in clause IV.9.2. That is to say, the RF-EMF of a 5G BS is measured during downlink traffic and no load is avoided.

During measurement, the instrument probe is placed on a support, the distance between the probe tip and the operator's trunk is ≥ 0.5 m the distance between the probe and 5G UE is maintained within the range 1 m to 3 m.

In addition, it is necessary to avoid or minimize the interference of other accidental EMF sources and the influence of the leakage current of the measuring instrument support.

IV.5 Measurement readings

The duration of each measurement point is ≥ 6 min, over which time the measured average value of the power density read from the instrument is the result.

According to the statistics of a large number of measurement cases, in most cases, the average power density reading on the instrument tends to be stable after continuous measurement for about 1 min. Therefore, a 1 min measurement time can be used when appropriate as a simple and feasible alternative to one of 6 min according to measurement needs.

NOTE – The integral value refers to the sum of the power density values contributed by all frequency points in the measured frequency band.

IV.6 Overall assessment

The overall electromagnetic environment quality index $EMQI$ is calculated according to the following equation:

$$EMQI = \frac{\sum_{i=1}^N (EMQI_{local})_i}{N} \quad (2)$$

In equation (2): $EMQI$ is the electromagnetic quality index of local area i ;

N : evaluates the number of local areas within an area.

Table IV.1 can be used to label the overall electromagnetic quality index level. The evaluation results can be used in the regional electromagnetic quality evaluation report and the regional electromagnetic quality index map.

Table IV.1 – Regional electromagnetic quality index

Electromagnetic quality index range (%)	$0 < EMQI \leq 20$	$20 < EMQI \leq 50$	$50 < EMQI \leq 100$	$100 < EMQI$
Electromagnetic quality index level	Level 1	Level 2	Level 3	Level 4

IV.7 Information record and reporting

IV.7.1 Information record for the grid method or spot measurements

Before undertaking measurements, basic information about the BS under test is collected, including the following:

- Physical site information: BS name, operator, construction site, latitude and longitude coordinates, transmission frequency range, type of antenna support, number of antennas, operation status, antenna height from the ground, etc.
- BS parameter information: collected according to monitoring purposes and needs, including transmission model, rated maximum power, actual transmitted power, antenna gain, average

load, antenna tilt angle (mechanical tilt angle plus electronic tilt angle), antenna lobe width (horizontal width, vertical width), antenna direction diagram, etc.

The same basic information is collected for BSs with other network systems at the same site.

While conducting measurements, to ensure the accuracy and traceability of measurement information, the following information is recorded:

- a) Relevant information about the BS under test collected as described above;
- b) Environmental temperature, relative humidity and weather conditions, as well as measurement date, start and end time, measurement personnel, etc;
- c) Measurement frequency range, instrument and probe model, including their calibration or verification certificate numbers, if any;
- d) The selected application scenarios, models and quantities of UE;
- e) A schematic diagram of the in-situ measuring points and the marked location of BS antenna, measurement points and other known electromagnetic radiation sources;
- f) The name (or latitude and longitude) of the measurement point, the vertical and horizontal distance between the measurement point and the BS transmitting antenna, the measurement data and a saved spectrum distribution map.

An example of the format of the in-situ measurement record and other forms are provided in clause IV.9 for reference.

IV.8 Measurement quality assurance

In order to ensure the reliability and accuracy of measurement data, the following quality assurance measures are taken:

- a) The selection of measurement points is representative and conforms to the specifications of clause IV.2.
- b) The measurement instrument (including probe) is certified or calibrated regularly and used within the validity period of the certificate. The instrument is also checked before and after measurement to ensure that the instrument is in normal working condition.
- c) Interference factors, including those due to humans and the environment, are excluded during measurement. The elimination of abnormal data and data processing of measurement results are handled according to statistical principles.
- d) A complete measurement file is established.

IV.9 Examples of forms

Examples of the forms used to record basic information and measurement results are shown as described in clauses IV.9.1–IV.9.4.

IV.9.1 Basic information record form

Table IV.2 is a basic information record form of the 5G in-situ RF exposure measurement.

Table IV.2 – Basic information record form

Basic information			
BS name		Operator	
Construction site		Transmitting frequency range	
Height from BS antenna to the ground		Type of antenna support	
Antenna number		Operating state	

Table IV.2 – Basic information record form

Measurement conditions			
Date and time of measurement		Model of measuring instrument	
Weather conditions		Measuring instrument number	
Environment temperature		Probe model	
Relative humidity		Probe number	
Measurement frequency range		Calibration certificate number of measuring instrument and probe	
BS EMF measurement point position diagram			
			

IV.9.2 Measurement result record form of the 5G in-situ RF exposure measurement

Table IV.3 is a measurement result record form of the 5G in-situ RF exposure measurement.

Table IV.3 – Measurement results record form

Measurement results									
S/N	Name of measurement point (or latitude and longitude)	Distance from the BS antenna		Application scenarios	Transmitting antenna		5G UE		Power density (W/m ²)
		Vertical	Horizontal		Operator	Downlink frequency band	Model	Quantity	
				☐ Data transmission ☐ Video interaction ☐ Game entertainment ☐ Virtual shopping ☐ Smart medical ☐ Industrial application ☐ Internet of vehicle ☐ Other_____					
				☐ Data transmission ☐ Video interaction ☐ Game entertainment ☐ Virtual shopping ☐ Smart medical ☐ Industrial application ☐ Internet of vehicle ☐ Other_____					
				☐ Data transmission ☐ Video interaction ☐ Game entertainment ☐ Virtual shopping ☐ Smart medical ☐ Industrial application ☐ Internet of vehicle ☐ Other_____					

IV.9.3 Measurement results record form for regional electromagnetic environment quality

Table IV.4 is a measurement results record form for regional electromagnetic environment quality.

Measurement task: _____ Measurement personnel: _____;

Measurement date: _____; Temperature: _____ ~ _____ °C; Humidity: _____ ~ _____ %;

Weather: _____; Instrument number: _____; Probe number: _____;

Measurement frequency band: _____; Resolution bandwidth: _____;
 Probe installation height: _____ m; Probe near to roof height: _____ m;
 Vehicle: _____.

Table IV.4 – Measurement results record form

Name of measurement point	Latitude	Longitude	Power density (S)	Power density Rate (%)	Measurement time
1					
2					
3					
4					
5					

IV.9.4 Assessment report of regional electromagnetic environment quality

Table IV.5 is the assessment report of regional electromagnetic environment quality.

Table IV.5 – Assessment report of regional electromagnetic environment quality

Assessment area	Urban built-up areas in _____ provinces, _____ cities, _____ counties, and _____ districts.
Assessment company	
Assessment conditions	Measurement date: _____; temperature: _____ ~ _____ °C; humidity: _____ ~ _____ %; weather: _____;
Assessment instrument and probe	Instrument number: _____; probe number: _____.
Assessment frequency	
Number of assessment points	Total number of effective measurement points: _____
Assessment results	1. Local area assessment results The local area corresponds to a grid edge length of _____ metres, totalling _____; Local distribution range of EMQI in local areas: _____ ~ _____; Local electromagnetic environment quality index level statistics: Level 1 _____, Level 2 _____, Level 3 _____, Level 4 _____. 2. Overall evaluation results Overall Electromagnetic EMQI = _____; Evaluate the level of electromagnetic environment quality index in the region: _____.

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